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Journal of Physics B: Atomic, Molecular and Optical Physics

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OPEN ACCESS

RECEIVED
8 February 2026

REVISED
22 March 2026

ACCEPTED FOR PUBLICATION
17 April 2026

PUBLISHED
4 May 2026

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PAPER

Accurate two-band tight-binding Hamiltonians for hexagonal two-dimensional materials for strong-field physics

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Keywords: HHG, ultrafast spectroscopy, strong-field physics, 2D materials

Abstract

Two-dimensional (2D) hexagonal materials are promising for a variety of applications ranging from valleytronics, spin-current sources, topology, and more. They are recently extensively studied for high-field nonlinear phenomena including Floquet physics, lightwave-driven photogalvanic currents, and high harmonic generation (HHG). For strong-field phenomena, which are typically induced by off-resonant intense driving lasers, two-band tight-binding (TB) descriptions of the system are highly popular because they provide a combination of computational efficiency and accuracy. They are especially appealing since they yield continuous analytic expressions for the band energies and dipole matrix elements, avoiding issues with gauge freedom that can arise in more complex *ab-initio* theory. However, implementations commonly utilize only up to 2nd nearest-neighbor (NN) terms, which can produce qualitatively wrong band structures in vast regions of the Brillouin zone (BZ). While this approximation is valid for low energy excitations near K/K', in strong-field conditions it can lead to wrong physical behavior. Here we develop accurate two-band TB Hamiltonians for hexagonal monolayers with up to 14 NN terms that are fitted to *ab-initio* bands in various systems (graphene, hexagonal-boron-nitride, WS_2, WSe_2, WTe_2, MoS_2, $MoSe_2$, $MoTe_2$), and reproduce the band structure across the entire BZ, including in regions with inverted curvature near Γ. We employ these models in semi-conductor Bloch equations simulations of HHG, showing that the angular-dependence of HHG yields substantially differs if choosing the full model, or a 2nd-order TB model that correctly captures bands only near K/K'. Our work provides accessible model parameters and analytic expressions for a variety of systems, which should facilitate accurate and efficient simulations of strong-field phenomena in 2D solids. It should also motivate more careful assessment of HHG features before applying various HHG spectroscopy techniques.

1. Introduction

Strong-field light-driven phenomena in solids have been intensely studied in recent years [1–4], especially in two-dimensional (2D) hexagonal materials. For instance, it has been shown that intense laser dressing allows tuning material properties in hexagonal boron-nitride (hBN) [5–7], or generating valley selective excitations in graphene [8, 9] and transition metal dichalcogenides (TMD) [10–13]. Strong-field driving was also shown to lead to unique Floquet physics in such systems [14–19], including excitonics [20] and Floquet topology [21]. Besides light-dressed states and valleytronics, 2D systems have also been heavily explored for high harmonic generation (HHG) [22–33] and bulk photogalvanic current generation schemes [8, 10, 20, 34, 35], which allow probing the system symmetries and band structure, ultrafast magnetism [36–41], coherent interference phenomena [42, 43], topology [44–47], and ultrafast correlations [28, 31].

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From a theoretical standpoint, ultrafast dynamics in 2D systems can be analyzed with a variety of techniques, either fully *ab-initio* through methods such as time-dependent density functional theory [48], or with model Hamiltonians in time-dependent Schrödinger equation [49] and semi-conductor Bloch equation (SBE) simulations [3, 50, 51]. Models are typically derived from *ab-initio* calculations of the ground state system and/or experiments in a variety of ways: (i) through wannerization [52–54], (ii) directly, by obtaining a discretized band model, (iii) by perturbation theory for a low-energy continuum model (e.g. $\mathbf{k}\cdot\mathbf{p}$), or (iv) by fitting tight-binding (TB) models [55–57]. Here we explore the latter approach since TB Hamiltonians provide a continuous expression for bands and transition dipole matrix elements (TDME), avoiding gauge freedom issues in the Bloch states, and since they can in principle be numerically exact while keeping numerical cost to a minimum. Most TB models employ two-band low-order expansion of nearest-neighbor (NN) terms. This obtains good bands near the K/K' valleys that are dominant features in 2D hexagonal solids, but fails to capture the correct band behavior throughout the Brillouin zone (BZ). In order to obtain good bands across the BZ multiple bands must be employed, and also higher order NN terms. However, specifically for strong-field phenomena, higher and lower lying bands contribute negligibly to dynamics due to the off-resonant driving. Moreover, upon incorporating additional bands exact analytic solutions are no longer guaranteed (while analytic solutions for even 3×3 or 4×4 Hamiltonians are available, they are very cumbersome, and higher-order cases are only solvable numerically), which also leads to issues in terms of gauge fixing. Further, multi-band models can make it difficult to obtain analytic insight into underlying physical mechanisms for strong-field processes such as HHG, and can substantially raise the numerical cost of simulations that typically scale parabolically with the number of bands (since the number of elements in the density matrix scales as $\propto N^2$ for $N\times N$ Hamiltonians). It is therefore desirable to maintain a 2×2 minimal Hamiltonian but still improve the band structure accuracy throughout the BZ, which is especially important for highly nonlinear phenomena where electrons are driven very far from K/K' valleys [5, 6, 43].

Here we develop an accurate 14th-order NN TB model for 2D hexagonal solids. By introducing an over-abundance of fitting parameters that respect the system's space group, this model artificially avoids introducing additional bands and keeps the Hamiltonian 2×2, but yields good bands throughout the BZ. Exact analytic expressions for the Hamiltonian, bands, wave functions, TDME, momentum matrix elements, and other quantities of interest, are provided. We perform a numerical fitting procedure for the model against density functional theory (DFT) simulations within the local density approximation for a variety of systems: graphene, hBN, WS_2, WSe_2, WTe_2, MoS_2, $MoSe_2$, $MoTe_2$. Tables with optimal NN terms for each system are provided. We employ these TB models for SBE simulations of strong-field driven HHG and explore the angular-dependence of HHG yields in standard experimental conditions. By comparing with simulations employing only 2nd-order NN TB models, we show that the HHG yields are sensitive to the band structure throughout the BZ (not only near K/K'), in particular in their angular dependence. This result motivates the application of improved band models for strong-field phenomena, especially considering that deviations/agreement between theory and experiments for such features is often used to study other physical effects such as dephasing [58] or evaluation the HHG mechanism itself [59, 60]. Our work should facilitate easy technical implementation for light-matter interactions simulations in hexagonal 2D solids that are accurate and feasible.

2. TB-Model derivation

Let us begin by formulating the TB model Hamiltonian and quantities of interest for strong-field simulations. While this task is a tedious technical one, it is at the heart of this paper and performing it rigorously here is meant to make future implementations straightforward. The lattice primitive vectors are given by $\mathbf{a}_1 = a_0\hat{\mathbf{x}}$, $\mathbf{a}_2 = a_0\left(-\hat{\mathbf{x}}/2+\sqrt{3}\hat{\mathbf{y}}/2\right)$, with a_0 the lattice parameter for the system. For the purposes of this derivation, we assume a simple honeycomb lattice structure as in graphene and hBN, with atomic sites situated at the origin (sublattice site A), and at $\mathbf{a}_1/3+2\mathbf{a}_2/3$ (sublattice site B). Note that this approach fails to account for the out-of-plane structure in TMDs, but in our case that is irrelevant because in 2D models out-of-plane dynamics are anyways neglected, and because we will soon introduce excessive free parameters to account for the correct band dispersion, artificially compensating for the loss of out-of-plane information.

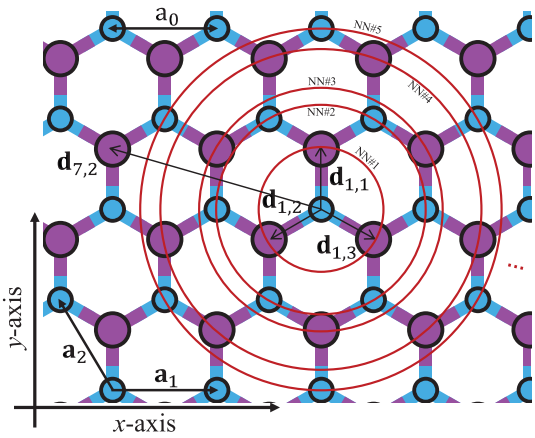


Figure 1. Schematic illustration of the honeycomb lattice and NN connections. Over the lattice there are indicators for the lattice vectors, lattice parameter, NN vectors, NN rings (up to 5th order), and $\boldsymbol{d}_{7,2}$ as an exemplary vector.

Figure 1 illustrates the lattice structure and resulting NN connection vectors. We can write out the NN vectors connecting AB sublattice sites, of which there are three:

$$\begin{aligned}\mathbf{d}_{1,1} &= \frac{a_0}{\sqrt{3}}\hat{\mathbf{y}} \\ \mathbf{d}_{1,2} &= \frac{a_0}{2}\left(-\hat{\mathbf{x}} - \frac{\hat{\mathbf{y}}}{\sqrt{3}}\right) \\ \mathbf{d}_{1,3} &= \frac{a_0}{2}\left(\hat{\mathbf{x}} - \frac{\hat{\mathbf{y}}}{\sqrt{3}}\right)\end{aligned} \tag{1}$$

These are conveniently denoted as $\mathbf{d}_{1j}$, with '1' indicating the 1st NN, and j the neighbor index. All other NN vectors beyond 1st-order can be expressed in terms of $\mathbf{d}_{1j}$ for convenience, and are denoted as $\mathbf{d}_{ij}$. Table A1 in the appendix expresses the NN vectors $\mathbf{d}_{ij}$ in terms of coefficients of $\mathbf{d}_{1j}$, and further indicates if the connection is between A–B sublattices or A–A sublattices. For instance, reading the table we find $\mathbf{d}_{53} = -1\mathbf{d}_{11} + 2\mathbf{d}_{12} - \mathbf{d}_{13}$, which is an A–A connection.

With the NN vectors we construct the structure factors that are defined as:

$$f_n(\mathbf{k}) = \sum_j \exp\left(i\mathbf{k}\cdot\mathbf{d}_{nj}\right) \tag{2}$$

where the sum in equation (2) runs over all of the NN sites for a given order n (due to rotational symmetry the number of sites is always a multiple of three). The Hamiltonian is constructed such that A–A structure factors contribute to the diagonal (in this case these are terms of order 2, 5, 6, 10, 11, and 14), and A–B factors contribute off-diagonally (terms of order 1, 3, 4, 7, 8, 9, 12, and 13). Additionally, a gap term (the gap at K/K' is taken to be a value Δ, indicating a mass term, or inversion symmetry breaking term) is added to a σ_z off-diagonal matrix. Overall, the Hamiltonian in the basis of A/B creation/annihilation operators takes the form:

$$H_0 = \epsilon_0\sigma_0 + \frac{\Delta}{2}\sigma_z + \begin{pmatrix} \alpha(\mathbf{k}) & \beta(\mathbf{k}) \\ \beta^*(\mathbf{k}) & \alpha(\mathbf{k}) \end{pmatrix} \tag{3}$$

where $\alpha(\mathbf{k})$, $\beta(\mathbf{k})$, and ϵ_0 are given as:

$$\begin{aligned}\alpha(\mathbf{k}) &= \begin{pmatrix} t_2 f_2(\mathbf{k}) + t_5 f_5(\mathbf{k}) + t_6 f_6(\mathbf{k}) \\ +t_{10} f_{10}(\mathbf{k}) + t_{11} f_{11}(\mathbf{k}) + t_{14} f_{14}(\mathbf{k}) \end{pmatrix} \\ \beta(\mathbf{k}) &= \begin{pmatrix} t_1 f_1(\mathbf{k}) + t_3 f_3(\mathbf{k}) + t_4 f_4(\mathbf{k}) + t_7 f_7(\mathbf{k}) \\ +t_8 f_8(\mathbf{k}) + t_9 f_9(\mathbf{k}) + t_{12} f_{12}(\mathbf{k}) + t_{13} f_{13}(\mathbf{k}) \end{pmatrix} \\ \epsilon_0 &= (3t_2 - 6t_5 + 3t_6 + 6t_{10} - 6t_{11} - 6t_{14} + \Delta/2)\end{aligned} \tag{4}$$

Here ϵ_0 is a trivial constant shift of the entire Hamiltonian by an identity matrix, σ_0, which is only added to conveniently shift the valence band (VB) edge at K/K' to zero energy. $\alpha(\mathbf{k})$ is a real-valued

function of $\mathbf{k}$, and $\beta(\mathbf{k})$ is complex-valued. Simplified functional expressions are provided in the appendix. It can be sometimes more straightforward to represent the Hamiltonian in the following form:

$$H_0 = B_0(\mathbf{k})\,\sigma_0 + \mathbf{B}(\mathbf{k})\cdot\boldsymbol{\sigma} \tag{5}$$

where $\boldsymbol{\sigma} = \{\sigma_x, \sigma_y, \sigma_z\}$ is a vector of Pauli matrices, and $B_0(\mathbf{k})$ and $\mathbf{B}(\mathbf{k})=\{B_x(\mathbf{k}), B_y(\mathbf{k}), B_z(\mathbf{k})\}$ can be extracted from the above definitions for $\alpha(\mathbf{k})$, $\beta(\mathbf{k})$, and ϵ_0:

$$\begin{aligned} B_0(\mathbf{k}) &= \alpha(\mathbf{k}) + \epsilon_0 \\ B_x(\mathbf{k}) &= \mathrm{Re}\{\beta(\mathbf{k})\} \\ B_y(\mathbf{k}) &= -\mathrm{Im}\{\beta(\mathbf{k})\} \\ B_z(\mathbf{k}) &= \Delta/2 \end{aligned} \tag{6}$$

where in equation (6) all of the functions are real-valued. Exact analytic expressions for the simplified functions of $\mathbf{B}(\mathbf{k})$ are provided in the appendix.

With this definition for H_0, we can diagonalize the Hamiltonian and obtain the eigenvalues corresponding to the VB: $\varepsilon_{\mathrm{VB}} = B_0 - |\mathbf{B}|$, and the conduction band (CB): $\varepsilon_{\mathrm{CB}} = B_0 + |\mathbf{B}|$. For brevity, we henceforth remove the $\mathbf{k}$ dependence of all functions, and define $|\mathbf{B}| \equiv \sqrt{B_x^2 + B_y^2 + B_z^2}$. The corresponding eigenfunctions (for the periodic part of the Bloch states) can be found in various gauges owing to Bloch gauge freedom (e.g. see examples in [44, 61]). We present two choices that we denote as gauge-I and gauge-II. We found gauge-I to be more convenient for gapped systems as it avoids issues with singularities at K/K' altogether:

$$\begin{aligned} |\psi_{\mathrm{I,VB}}\rangle &= \frac{1}{N_{\mathrm{I}}}\left\{iB_y - B_x,\ B_z + |\mathbf{B}|\right\} \\ |\psi_{\mathrm{I,CB}}\rangle &= \frac{1}{N_{\mathrm{I}}}\left\{B_z + |\mathbf{B}|,\ B_x + iB_y\right\} \\ N_{\mathrm{I}} &= \sqrt{(B_z + |\mathbf{B}|)^2 + B_x^2 + B_y^2} \end{aligned} \tag{7}$$

In gauge-I the normalization constant N_{I} is positive definite as long as $\Delta > 0$, even at K/K' points for which we have $B_x(\mathrm{K}) = B_y(\mathrm{K}) = 0$. Alternatively, the states can be expressed in gauge-II:

$$\begin{aligned} |\psi_{\mathrm{II,VB}}\rangle &= \frac{1}{N_{\mathrm{II}}}\left\{1,\ -\frac{B_x + iB_y}{|\mathbf{B}| - B_z}\right\} \\ |\psi_{\mathrm{II,CB}}\rangle &= \frac{1}{N_{\mathrm{II}}}\left\{1,\ \frac{B_x + iB_y}{|\mathbf{B}| + B_z}\right\} \\ N_{\mathrm{II}} &= \sqrt{1 + \frac{B_x^2 + B_y^2}{\sqrt{|\mathbf{B}|^2 + B_z^2}}} \end{aligned} \tag{8}$$

In gauge-II, even for $\Delta = 0$, we have that $N_{\mathrm{II}} = \sqrt{2}$ and the states do not diverge, unlike in gauge-I (because numerators and denominators diverge at the same rate). However, gauge-II can be more numerically challenging around K/K' points for the case of $\Delta > 0$. As we will show below, for the specific case of $\Delta = 0$ (e.g. for graphene-type systems) the TDME in gauge-I and-II both diverge, but the divergence in gauge-II is numerically softer.

Next, we fit the hopping amplitudes to match model bands to DFT band structures obtained in the local density approximation and neglecting spin degrees of freedom (obtained using octopus code [62]). This approach can be well suited for many strong-field phenomena, including valleytronics, but might fail if spin currents, magnetization, or topology, are of interest. Generally, spin-orbit coupling terms of various forms can be added to the Hamiltonian in order to capture such phenomena and the small spin-split bands in TMDs, but this is beyond our scope here (since adding those will effectively turn the Hamiltonian into 4×4 form, and also those are usually negligible in TMDs apart from some impact in weak light-induced magnetic states [37–41]). The value of Δ represents the gap at K and is taken directly from DFT bands, while the hopping amplitudes t_j are free parameters for the following cost function:

$$g(\mathbf{t}) = \frac{1}{N_k}\sum_{k_i \in BZ} w_k \left(\frac{1}{2}\left|\varepsilon_{\mathrm{VB}}(k_i) - \varepsilon_{\mathrm{VB}}^{\mathrm{DFT}}(k_i)\right| + \frac{1}{2}\left|\varepsilon_{\mathrm{CB}}(k_i) - \varepsilon_{\mathrm{CB}}^{\mathrm{DFT}}(k_i)\right|\right) \tag{9}$$

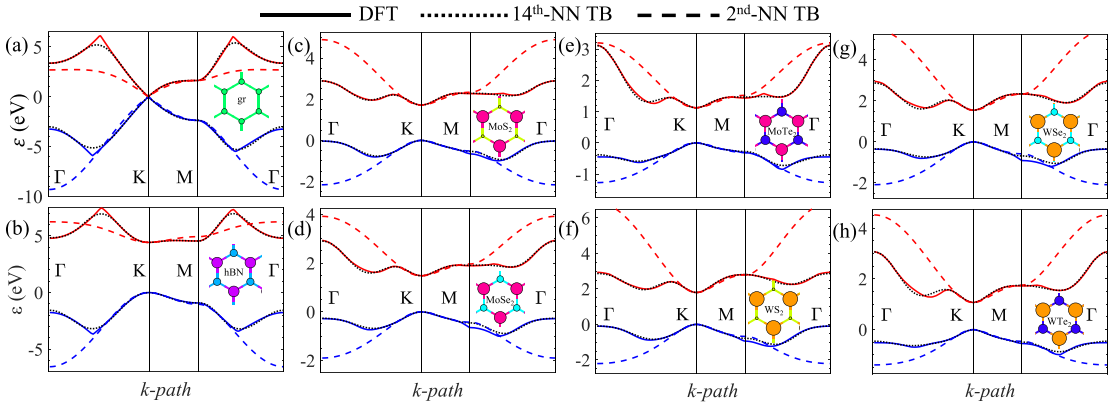


Figure 2. Optimized TB band structures along high symmetry lines at the 14th NN level (dotted black line) compared to *ab-initio* DFT bands (solid lines), and to optimized simplified 2nd NN TB model (dashed lines). The material systems are indicated in insets.

where we denote $\mathbf{t} \equiv \{t_1, t_2, \ldots t_{14}\}$ as a vector of hopping amplitudes, and where $\varepsilon^{\mathrm{DFT}}_{\mathrm{VB/CB}}$ are the DFT Kohn-Sham band eigenenergies. The sum in equation (9) runs over the BZ with k_i being numerically sampled k-points on the grid, w_k the weight within the cost function for a given k-point, and N_k is the number of k-points in the grid. For optimally fitting the 14th NN model we used a uniform Γ-centered k-grid in the first BZ, but set k-points near the K/K' valleys (upholding the condition $k_1 + k_2 > 0.5$, where k_1 and k_2 are fractional coordinates in units of the reciprocal lattice vectors, $\mathbf{b}_1$, $\mathbf{b}_2$) with double weight. This enforces that the most important fitting regions are the K/K' valleys. For comparison purposes in later sections, we also employed a simplified approach by allowing only up to 2nd NN terms (setting higher order terms in the cost function to zero), while fitting the hopping amplitudes only near the K/K' valleys and ignoring the rest of the BZ.

Results for TB bands obtained within this 14 NN approach can be seen for all sample materials in figure 2. As seen, the band structures agree very well throughout the BZ, including additional oscillatory patterns across $\Gamma - K$ and Γ -M that are common in TMDs. Such features are usually neglected in TB models because they can be very difficult to capture and usually require additional on-site orbitals in the model (more bands). At this point we highlight that physically, the matching for some of the band features is artificial in the sense that we enforce them by over-fitting long-range hopping—such long-range hopping terms are certainly physically negligible in insulating materials. However, this brute-force approach allows obtaining proper band energies even while ignoring band hybridization, thus keeping the Hamiltonian at a minimal two-band size. In practical light–matter interaction calculations the band energies play a crucial role, far greater than the orbital character of various regions in the BZ (which is anyways often neglected). In that respect, this over-fitted model provides an easy approach to test physical phenomena with efficient compact Hamiltonians.

At a next stage we obtain the TDME and momentum matrix elements, which will be employed in the SBE simulations below. The TDME can be analytically obtained by:

$$\begin{aligned}
\mathbf{d}_{\mathbf{cv}}(\mathbf{k}) &= i\left\langle \psi_{\mathrm{CB}}(\mathbf{k}) \middle| \frac{\partial}{\partial \mathbf{k}} \middle| \psi_{\mathrm{VB}}(\mathbf{k}) \right\rangle \\
\mathbf{d}_{\mathbf{vv}}(\mathbf{k}) &= i\left\langle \psi_{\mathrm{VB}}(\mathbf{k}) \middle| \frac{\partial}{\partial \mathbf{k}} \middle| \psi_{\mathrm{VB}}(\mathbf{k}) \right\rangle \\
\mathbf{d}_{\mathbf{cc}}(\mathbf{k}) &= i\left\langle \psi_{\mathrm{CB}}(\mathbf{k}) \middle| \frac{\partial}{\partial \mathbf{k}} \middle| \psi_{\mathrm{CB}}(\mathbf{k}) \right\rangle
\end{aligned} \tag{10}$$

and are gauge-dependent. Explicit simplified expressions are given in the appendix in both gauges. Similarly, we can obtain the momentum matrix elements directly and analytically as:

$$\begin{aligned}
\mathbf{p}_{\mathbf{cv}}(\mathbf{k}) &= \left\langle \psi_{\mathrm{CB}}(\mathbf{k}) \middle| \frac{\partial H_0(\mathbf{k})}{\partial \mathbf{k}} \middle| \psi_{\mathrm{VB}}(\mathbf{k}) \right\rangle \\
\mathbf{p}_{\mathbf{vv}}(\mathbf{k}) &= \left\langle \psi_{\mathrm{VB}}(\mathbf{k}) \middle| \frac{\partial H_0(\mathbf{k})}{\partial \mathbf{k}} \middle| \psi_{\mathrm{VB}}(\mathbf{k}) \right\rangle. \\
\mathbf{p}_{\mathbf{cc}}(\mathbf{k}) &= \left\langle \psi_{\mathrm{CB}}(\mathbf{k}) \middle| \frac{\partial H_0(\mathbf{k})}{\partial \mathbf{k}} \middle| \psi_{\mathrm{CB}}(\mathbf{k}) \right\rangle
\end{aligned} \tag{11}$$

Note that these expressions for momentum matrix elements are accurate only for this two-band model where there is a single orbital per lattice site, implying intra-site transitions are negligible (in our case non-existent) [63]. Lastly, the gauge-invariant Berry curvature (out-of-plane z-component) is obtained through:

$$\begin{aligned}\Omega_{\mathrm{CB}}(\mathbf{k}) &= i\left(\left\langle \frac{\partial}{\partial k_x}\psi_{\mathrm{CB}}(\mathbf{k})\middle|\frac{\partial}{\partial k_y}\psi_{\mathrm{CB}}(\mathbf{k})\right\rangle - \left\langle \frac{\partial}{\partial k_y}\psi_{\mathrm{CB}}(\mathbf{k})\middle|\frac{\partial}{\partial k_x}\psi_{\mathrm{CB}}(\mathbf{k})\right\rangle\right)\\ \Omega_{\mathrm{VB}}(\mathbf{k}) &= i\left(\left\langle \frac{\partial}{\partial k_x}\psi_{\mathrm{VB}}(\mathbf{k})\middle|\frac{\partial}{\partial k_y}\psi_{\mathrm{VB}}(\mathbf{k})\right\rangle - \left\langle \frac{\partial}{\partial k_y}\psi_{\mathrm{VB}}(\mathbf{k})\middle|\frac{\partial}{\partial k_x}\psi_{\mathrm{VB}}(\mathbf{k})\right\rangle\right)\end{aligned}. \tag{12}$$

For $\Delta = 0$ one obtains $\Omega_{\mathrm{VB/CB}} = 0$, as should be the case for inversion-symmetric systems.

3. HHG With SBEs

We now employ the TB Hamiltonian developed above for HHG simulations in 2D hexagonal solids. We apply the SBE approach in the density matrix formalism with phenomenological dephasing, in the length gauge, and in the instantaneous Houston basis [3]. The equations of motion for the density matrix in atomic units take the form:

$$\begin{aligned}\frac{\partial}{\partial t}\rho_{vv}(\mathbf{k},t) &= i\mathbf{E}(t)\cdot\left[\mathbf{d_{cv}}(\mathbf{k})\,\rho^{*}_{cv}(\mathbf{k},t) - \mathbf{d^{*}_{cv}}\,\rho_{cv}(\mathbf{k},t)\right]\\ \frac{\partial}{\partial t}\rho_{cv}(\mathbf{k},t) &= -i\left[\begin{array}{c}\left(\varepsilon_{CB}(\mathbf{k}(t)) - \varepsilon_{VB}(\mathbf{k}(t)) - \frac{i}{T_2}\right)\rho_{cv}(\mathbf{k},t)\\ +E(t)\cdot\left(\begin{array}{c}(\mathbf{d_{cc}}(\mathbf{k}) - \mathbf{d_{vv}}(\mathbf{k}))\,\rho_{cv}(\mathbf{k},t)\\ +\mathbf{d_{cv}}(\mathbf{k})\,(2\rho_{vv}(\mathbf{k},t) - 1)\end{array}\right)\end{array}\right]\end{aligned} \tag{13}$$

where the time-dependent k-vector is given by $\mathbf{k}(t) = \mathbf{k}_0 + \frac{1}{c}\mathbf{A}(t)$, with $\mathbf{k}_0$ the crystal momentum at $t = 0$, and $\mathbf{E}(t)$ the electric field vector (in the dipole approximation), which is connected to the vector potential via: $-\partial_t\mathbf{A}(t) = c\mathbf{E}(t)$, and c is the speed of light. In equation (13) we use the relations $\rho_{\mathrm{cc}} = 1 - \rho_{\mathrm{vv}}$, $\rho_{\mathrm{vc}} = \rho^{*}_{\mathrm{cv}}$, and the initial conditions $\rho_{\mathrm{cv}}(t=0) = \rho_{\mathrm{cc}}(t=0) = 0$. T_2 is the interband coherence dephasing time that accounts for electron–electron and electron–phonon induced dephasing of the off-diagonal density-matrix terms (note though that this does not include all effects of electron–phonon coupling as occupations are not allowed to transition between different k-points [64], and the dephasing time is k-independent [65]).

The SBE are numerically solved using a 4th order Runge–Kutta approach using a time step of 0.48 as, and employing the following electric field vector:

$$\mathbf{E}(t) = f(t)\,E_0\left[\cos(\theta)\,\hat{\mathbf{x}} + \sin(\theta)\,\hat{\mathbf{y}}\right]\cos(\omega t) \tag{14}$$

where θ is the driving angle (which is varied in simulations), $f(t)$ is a temporal envelope function taken to have a super-sin form [66] (similar to super Gaussian) with a full-width-half-max of 21.3 fs, ω corresponds to a driving wavelength of 1600 nm, and we use moderately intense driving power in the strong-field regime corresponding to 5×10^{11} W cm^{-2}. Note that the results obtained here should not depend on the envelope function choice, which is used merely for numerical convenience. For gapped TMDs and monolayer hBN the SBE are solved in gauge-I with a 120 × 120 Γ-centered k-grid, while for graphene we employ gauge-II with a denser 360 × 360 -centered k-grid (which is required to obtain finer sampling around the Dirac cone). We calculate the time-dependent electronic current in the system:

$$\mathbf{J}(t) = \mathbf{J_{intra}}(t) + \mathbf{J_{inter}}(t) \tag{15}$$

where the interband and intraband currents have the form:

$$\begin{aligned}\mathbf{J_{intra}}(t) &= -\sum_{\mathbf{k}\in BZ}\left[\rho_{vv}(\mathbf{k},t)\,\mathbf{p_{vv}}(\mathbf{k}(t)) + \rho_{cc}(\mathbf{k},t)\,\mathbf{p_{cc}}(\mathbf{k}(t))\right]\\ \mathbf{J_{inter}}(t) &= -\sum_{\mathbf{k}\in BZ} 2\mathrm{Re}\left[\rho_{cv}(\mathbf{k},t)\,\mathbf{p_{vc}}(\mathbf{k}(t))\right].\end{aligned} \tag{16}$$

From the current we obtain the HHG spectra as $I_{\mathrm{HHG}}(\omega_0) = \left|\int \mathrm{d}t f(t)\,\frac{\partial \mathbf{J}(t)}{\partial t}\mathrm{e}^{-\mathrm{i}\omega_0 t}\right|^2$.

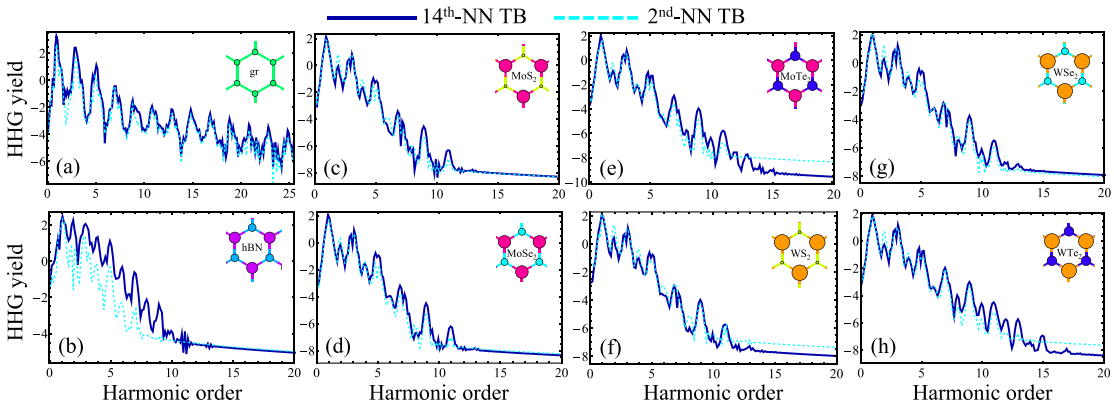


Figure 3. HHG spectra obtained from the 14NN TB model (solid lines) and the 2NN TB model (dashed lines) driven at $\theta = 0$. The spectra are presented in log scale. Clear quantitative deviations can be seen (e.g. at the 2nd harmonic for TMDs, general structure in hBN, and harmonic widths in graphene).

We apply this scheme to calculate the angle-resolved HHG yields for all sample material systems, which are of main interest in various HHG-based ultrafast spectroscopies [26, 47, 58, 67–73]. For this we employ two levels of TB models: (i) the optimized 14 NN model (solid lines in figures 2–4), and (ii) a minimal 2nd-order TB model that is optimally fitted only around K/K' valleys (dashed lines in figures 2–4). Figure 3 presents exemplary spectra driven at $\theta = 0$ (a high-symmetry axis) in all systems, demonstrating that already at the level of total HHG yields and the general structure of the HHG spectrum, the improved TB model makes a large difference. In other words, because the driving field is so intense (as usually employed for HHG), electrons are driven to regions sufficiently far from the K/K' valleys, meaning that if our model Hamiltonian does not correctly capture the band structure in those regions the results will be quantitatively wrong. This can be seen in multiple quantitative features such as the harmonic widths in graphene, the HHG decay rate in hBN, large yield deviations in even harmonic values across TMDs (especially 2nd harmonic yields), etc. We expect that such features are condition-dependent, and should worsen if increasing the driving intensity or wavelength (yielding an overall larger vector potential and electronic excursion in the BZ). Notably, this model was recently employed to show that the qualitative behavior of the HHG indeed can depend on electron occupation throughout the BZ (even far from K/K' regions) in agreement with experiments in WS_2 (with features that could not be captured by a 2nd-order NN TB model) [43], as well as in graphite [74]. This validates the approach and its potential significance in general HHG settings.

Figure 4 further presents harmonic-orders angular dependence for one select harmonic from each material system that shows a strong qualitative deviation between the models (full data sets are delegated to the appendix). We obtain the normalized angle-dependent yields by integrating over the HHG yield around the harmonic peaks and normalizing to the maximal value. We again observe clear deviations arising between the HHG emission calculated from both models, but not only quantitatively, also in the qualitative HHG behavior. For example, this can lead to the wrong maximizing yield angle in $MoTe_2$ in the 7th harmonic, or lead to a weak angular dependence in the 5th harmonic in MoS_2. Other deviations could include overall wrong qualitative shape for the angular dependence such as in the 9th harmonic in graphene, or 10th harmonic in $MoSe_2$. We should emphasize though that the majority of harmonics behave similarly between models and are not so sensitive to the electronic structure beyond the K/K' regions. Nonetheless, for every system at least one strong signature of these band deviations arose, and more are expected to appear if using larger vector potentials. Such features can arise for particular harmonic orders that are resonant in several regions in the BZ that interfere, depending on the laser and material parameters (and the interference details strongly depend on the shape of the bands, thus being sensitive to higher order hopping terms). Overall, our simulations reveal that for proper modeling of HHG and other strong-field phenomena one must first obtain sufficiently good ground state electronic structure, including in regions farther out from K/K' for hexagonal systems (where the results should be transferable also to other solids).

Lastly, for monolayer hBN we also explore the T_2-dependence of the angular-dependent HHG yields. Different studies usually employ varying values for T_2 in hBN, often in an attempt to obtain better agreement between theory and experiments. For example, reference [7] recently employed a value of $T_2 = 6$ (fs). However, recent studies found the *ab-initio* and experimental coherence times closer to ∼25 fs [64]. Figure 5 presents the angular dependence of HHG yields, similar to figure 3(b), but

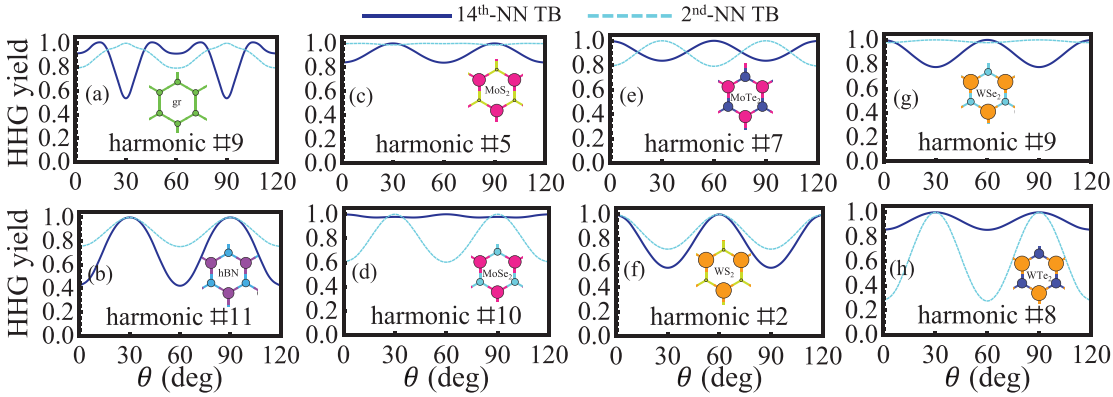


Figure 4. Angle-resolved normalized harmonic yields, per select harmonic orders (indicated in the figure along with the material system) that show strong deviations between the 14NN TB model (solid lines) and the 2NN TB model (dashed lines), in each material system. Full data sets are provided in the appendix. In some cases very strong qualitative changes can be observed in the HHG angular dependence for specific harmonic orders depending on the level of TB model.

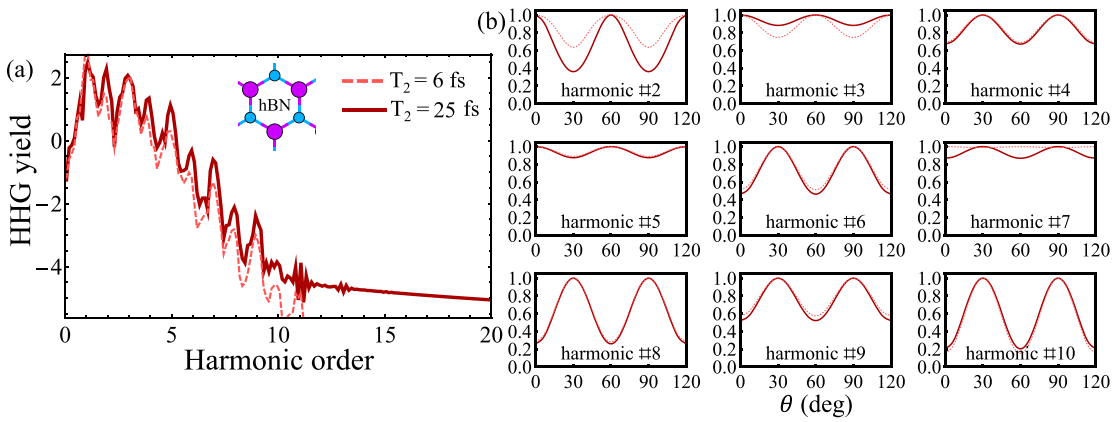


Figure 5. (a) HHG spectra at $\theta = 0$ for hBN with the 14th-NN optimized TB model, using different values of T_2: 25 fs (solid lines) and 6 (fs) (dashed lines). (b) Corresponding normalized harmonic yield angular dependence. The qualitative features are roughly independent of T_2, except for the 7th harmonic that shows a strong sensitivity to T_2.

comparing the two cases of $T_2 = 6$ or $T_2 = 25$ (fs) (for the 14NN accurate TB model). The results show that while there is a quantitative deviation between the simulations, almost all angular-dependent qualitative HHG features are the same. The only exception in this case is the 7th harmonic, which is qualitatively different and shows a strong T_2-dependence. In that respect, our results urge care in analyzing specific harmonic features given that phenomenological T_2 dephasing times anyways do not account for the correct BZ phonon-induced scattering effects. This is especially relevant if strong features agreeing with experiments are used to extrapolate to other behavior observed in simulations.

4. Summary

To summarize, we put forward a long-range TB model for hexagonal 2D systems, optimizing TB parameters to *ab-initio* bands. By incorporating abundant degrees of freedom and free parameters, we avoided introducing additional bands that are typically negligible in strong-field phenomena, keeping the Hamiltonian compact. Importantly, this permits good band structures throughout the BZ, including near Γ and M points, which might become relevant in strong-field and ultrafast phenomena. We further employed this accurate TB model for HHG calculations with SBE in various 2D hexagonal solids, showing that spectra are generally sensitive to the level of model even beyond the K/K' region. The sensitivity manifests both quantitatively in HHG yields, but also qualitatively in behavior of angle-dependent HHG yields for select harmonic orders. We also showed that T_2-dependence can lead to similar sensitivities and should be chosen with care. Our work provides a simple numerical scheme to model strong-field phenomena in hexagonal systems avoiding multiple bands and complex *ab-initio* simulations, and should find practical use as the field evolves.

Acknowledgments

ON gratefully acknowledges the scientific support of Prof. Dr. Angel Rubio, as well as fruitful discussions with Prof. Dieter Bauer, Prof. Marcelo Ciappinna, Prof. Gopal Dixit, and Prof. Lun Yue. ON also gratefully acknowledges support from the Technion Helen Diller Quantum Center and RBNI programs Nevet grants.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix A. Technical details

We describe here additional technical and analytical details discussed in the main text, such as equations for underlying functions, transition dipole matrix elements (TDME) expressions, vectors for NN sites, etc. First, table 1 describes the high-order NN vectors written in the basis of the $\mathbf{d}_{1j}$ (the 1st-order NN vectors).

Table 1. Higher-order NN vectors up to 14th order in the basis of $\mathbf{d}_{1j}$. For each connection order the table also indicates if it connects A–A or A–B sublattices.

	d_{11}	d_{12}	d_{13}		d_{11}	d_{12}	d_{13}		d_{11}	d_{12}	d_{13}
$\boldsymbol{d_{21}}$ **(A–A)**	−1	1	0	$\boldsymbol{d_{71}}$ **(A–B)**	2	−2	1	$\boldsymbol{d_{11,1}}$ **(A–A)**	3	−3	0
$\boldsymbol{d_{22}}$	1	−1	0	$\boldsymbol{d_{72}}$	1	2	−2	$\boldsymbol{d_{11,2}}$	0	3	−3
$\boldsymbol{d_{23}}$	0	1	−1	$\boldsymbol{d_{73}}$	−2	1	2	$\boldsymbol{d_{11,3}}$	−3	0	3
$\boldsymbol{d_{24}}$		−1	1	$\boldsymbol{d_{74}}$	2	1	−2	$\boldsymbol{d_{11,4}}$	3	0	−3
$\boldsymbol{d_{25}}$	1	0	−1	$\boldsymbol{d_{75}}$	−2	2	1	$\boldsymbol{d_{11,5}}$	−3	3	0
$\boldsymbol{d_{26}}$	−1	0	1	$\boldsymbol{d_{76}}$	1	−2	2	$\boldsymbol{d_{11,6}}$	0	−3	3
$\boldsymbol{d_{31}}$ **(A-B)**	−2	0	0	$\boldsymbol{d_{81}}$ **(A–B)**	3	−1	−1	$\boldsymbol{d_{12,1}}$	3	−3	1
$\boldsymbol{d_{32}}$	0	−2	0	$\boldsymbol{d_{82}}$	−1	3	−1	$\boldsymbol{d_{12,2}}$ **(A–B)**	1	3	−3
$\boldsymbol{d_{33}}$	0	0	−2	$\boldsymbol{d_{83}}$	−1	−1	3	$\boldsymbol{d_{12,3}}$	−3	1	3
$\boldsymbol{d_{41}}$ **(A–B)**	2	−1	0	$\boldsymbol{d_{91}}$ **(A–B)**	3	−2	0	$\boldsymbol{d_{12,4}}$	3	1	−3
$\boldsymbol{d_{42}}$	0	2	−1	$\boldsymbol{d_{92}}$	0	3	−2	$\boldsymbol{d_{12,5}}$	−3	3	1
$\boldsymbol{d_{43}}$	−1	0	2	$\boldsymbol{d_{93}}$	−2	0	3	$\boldsymbol{d_{12,6}}$	1	−3	3
$\boldsymbol{d_{44}}$	2	0	−1	$\boldsymbol{d_{94}}$	3	0	−2	$\boldsymbol{d_{13,1}}$	4	−2	−1
$\boldsymbol{d_{45}}$	−1	2	0	$\boldsymbol{d_{95}}$	−2	3	0	$\boldsymbol{d_{13,2}}$	−1	4	−2
$\boldsymbol{d_{46}}$	0	−1	2	$\boldsymbol{d_{96}}$	0	−2	3	$\boldsymbol{d_{13,3}}$	−2	−1	4
$\boldsymbol{d_{51}}$ **(A–A)**	2	−1	−1	$\boldsymbol{d_{10,1}}$ **(A–A)**	3	−2	−1	$\boldsymbol{d_{13,4}}$	4	−1	−2
$\boldsymbol{d_{52}}$	−2	1	1	$\boldsymbol{d_{10,2}}$	−1	3	−2	$\boldsymbol{d_{13,5}}$	−2	4	−1
$\boldsymbol{d_{53}}$	−1	2	−1	$\boldsymbol{d_{10,3}}$	−2	−1	3	$\boldsymbol{d_{13,6}}$	−1	−2	4
$\boldsymbol{d_{54}}$	1	−2	1	$\boldsymbol{d_{10,4}}$	3	−1	−2	$\boldsymbol{d_{14,1}}$	2	2	−4
$\boldsymbol{d_{55}}$	−1	−1	2	$\boldsymbol{d_{10,5}}$	−2	3	−1	$\boldsymbol{d_{14,2}}$	−4	2	2
$\boldsymbol{d_{56}}$	1	1	−2	$\boldsymbol{d_{10,6}}$	−1	−2	3	$\boldsymbol{d_{14,3}}$	2	−4	2
$\boldsymbol{d_{61}}$ **(A–A)**	2	−2	0	$\boldsymbol{d_{10,7}}$	−3	2	1	$\boldsymbol{d_{14,4}}$	−2	−2	4
$\boldsymbol{d_{62}}$	0	2	−2	$\boldsymbol{d_{10,8}}$	1	−3	2	$\boldsymbol{d_{14,5}}$	4	−2	−2
$\boldsymbol{d_{63}}$	−2	0	2	$\boldsymbol{d_{10,9}}$	2	1	−3	$\boldsymbol{d_{14,6}}$	−2	4	−2
$\boldsymbol{d_{64}}$	2	0	−2	$\boldsymbol{d_{10,10}}$	−3	1	2				
$\boldsymbol{d_{65}}$	−2	2	0	$\boldsymbol{d_{10,11}}$	2	−3	1				
$\boldsymbol{d_{66}}$	0	−2	2	$\boldsymbol{d_{10,12}}$	1	2	−3				

We next explicitly write the simplified mathematical forms of various functions and expressions employed in the main text and numerical simulations. The form of $\alpha(\mathbf{k})$:

$$\alpha(\mathbf{k}) = 2\begin{pmatrix} \left(C_x + C_{\frac{1}{2}x-\frac{\sqrt{3}}{2}y} + C_{\frac{1}{2}x-\frac{\sqrt{3}}{2}y}\right)t_2 + \left(C_{\sqrt{3}y} + C_{\frac{3}{2}x-\frac{\sqrt{3}}{2}y} + C_{\frac{3}{2}x+\frac{\sqrt{3}}{2}y}\right)t_5 \\ + \left(C_{2x} + C_{x-\sqrt{3}y} + C_{x+\sqrt{3}y}\right)t_6 \\ + \left(C_{\frac{1}{2}x-\frac{3\sqrt{3}}{2}y} + C_{2x-\sqrt{3}y} + C_{\frac{5}{2}x-\frac{\sqrt{3}}{2}y} + C_{2x+\sqrt{3}y} + C_{\frac{5}{2}x+\frac{\sqrt{3}}{2}y} + C_{\frac{1}{2}x+\frac{3\sqrt{3}}{2}y}\right)t_{10} \\ + \left(C_{3x} + C_{\frac{3}{2}x-\frac{3\sqrt{3}}{2}y} + C_{\frac{3}{2}x+\frac{3\sqrt{3}}{2}y}\right)t_{11} + \left(C_{2\sqrt{3}y} + C_{3x-\sqrt{3}y} + C_{3x+\sqrt{3}y}\right)t_{14} \end{pmatrix} \tag{17}$$

with the shortened notation:

$$C_{ax+by}=\cos\left(ak_x+bk_y\right),S_{ax+by}=\sin\left(ak_x+bk_y\right). \tag{18}$$

The form for $B_x(\mathbf{k})$, $B_y(\mathbf{k})$ (from which one can obtain the form for $\beta(\mathbf{k})$):

$$B_x(\mathbf{k})=\begin{pmatrix}\left(C_{\frac{y}{\sqrt{3}}}+C_{\frac{x}{2}-\frac{\sqrt{3}y}{6}}+C_{\frac{x}{2}+\frac{\sqrt{3}y}{6}}\right)t_1+\left(C_{\frac{2y}{\sqrt{3}}}+C_{x-\frac{y}{\sqrt{3}}}+C_{x+\frac{y}{\sqrt{3}}}\right)t_3\\+\left(C_{x-\frac{2y}{\sqrt{3}}}+C_{x+\frac{2y}{\sqrt{3}}}+C_{\frac{x}{2}+\frac{5\sqrt{3}}{6}y}+C_{\frac{x}{2}-\frac{5\sqrt{3}}{6}y}+C_{\frac{3x}{2}-\frac{\sqrt{3}y}{6}}+C_{\frac{3x}{2}+\frac{\sqrt{3}y}{6}}\right)t_4\\+\left(C_{2x-\frac{y}{\sqrt{3}}}+C_{2x+\frac{y}{\sqrt{3}}}+C_{\frac{x}{2}-\frac{7\sqrt{3}}{6}y}+C_{\frac{x}{2}+\frac{7\sqrt{3}}{6}y}+C_{\frac{3x}{2}-\frac{5\sqrt{3}}{6}y}+C_{\frac{3x}{2}+\frac{5\sqrt{3}}{6}y}\right)t_7\\+\left(C_{\frac{4y}{\sqrt{3}}}+C_{2x-\frac{2y}{\sqrt{3}}}+C_{2x+\frac{2y}{\sqrt{3}}}\right)t_8\\+\left(C_{x-\frac{4y}{\sqrt{3}}}+C_{x+\frac{4y}{\sqrt{3}}}+C_{\frac{3x}{2}-\frac{7\sqrt{3}}{6}y}+C_{\frac{3x}{2}+\frac{7\sqrt{3}}{6}y}+C_{\frac{5x}{2}-\frac{\sqrt{3}y}{6}}+C_{\frac{5x}{2}+\frac{\sqrt{3}y}{6}}\right)t_9\\+\left(C_{2x+\frac{4y}{\sqrt{3}}}+C_{2x-\frac{4y}{\sqrt{3}}}+C_{x+\frac{5y}{\sqrt{3}}}+C_{x-\frac{5y}{\sqrt{3}}}+C_{3x-\frac{y}{\sqrt{3}}}+C_{3x+\frac{y}{\sqrt{3}}}\right)t_{12}\\+\left(C_{3x-\frac{2y}{\sqrt{3}}}+C_{3x+\frac{2y}{\sqrt{3}}}+C_{\frac{x}{2}-\frac{11\sqrt{3}}{6}y}+C_{\frac{x}{2}+\frac{11\sqrt{3}}{6}y}+C_{\frac{5x}{2}-\frac{7\sqrt{3}}{6}y}+C_{\frac{5x}{2}+\frac{7\sqrt{3}}{6}y}\right)t_{13}\end{pmatrix} \tag{19}$$

and

$$B_y(\mathbf{k})=\begin{pmatrix}\left(S_{\frac{x}{2}+\frac{\sqrt{3}}{6}}-S_{\frac{x}{2}-\frac{\sqrt{3}y}{6}}-S_{\frac{y}{\sqrt{3}}}\right)t_1+\left(S_{\frac{2y}{\sqrt{3}}}+S_{x-\frac{y}{\sqrt{3}}}-S_{x+\frac{y}{\sqrt{3}}}\right)t_3\\+\left(S_{x+\frac{2y}{\sqrt{3}}}-S_{x-\frac{2y}{\sqrt{3}}}+S_{\frac{x}{2}-\frac{5\sqrt{3}}{6}y}-S_{\frac{x}{2}+\frac{5\sqrt{3}}{6}y}+S_{\frac{3x}{2}\frac{+\sqrt{3}y}{6}}-S_{\frac{3x}{2}-\frac{\sqrt{3}y}{6}}\right)t_4\\+\left(S_{2x-\frac{y}{\sqrt{3}}}-S_{2x+\frac{y}{\sqrt{3}}}+S_{\frac{x}{2}+\frac{7\sqrt{3}}{6}y}-S_{\frac{x}{2}-\frac{7\sqrt{3}}{6}y}+S_{\frac{3x}{2}-\frac{5\sqrt{3}}{6}y}-S_{\frac{3x}{2}+\frac{5\sqrt{3}}{6}y}\right)t_7\\+\left(S_{2x+\frac{2y}{\sqrt{3}}}-S_{2x-\frac{2y}{\sqrt{3}}}-S_{\frac{4y}{\sqrt{3}}}\right)t_8\\+\left(S_{x-\frac{4y}{\sqrt{3}}}-S_{x+\frac{4y}{\sqrt{3}}}+S_{\frac{3x}{2}+\frac{7\sqrt{3}}{6}y}-S_{\frac{3x}{2}-\frac{7\sqrt{3}}{6}y}+S_{\frac{5x}{2}+\frac{\sqrt{3}}{6}y}-S_{\frac{5x}{2}-\frac{\sqrt{3}}{6}y}\right)t_9\\+\left(S_{x+\frac{5}{\sqrt{3}}y}-S_{x-\frac{5}{\sqrt{3}}y}+S_{2x-\frac{4y}{\sqrt{3}}}-S_{2x+\frac{4y}{\sqrt{3}}}+S_{3x-\frac{y}{\sqrt{3}}}-S_{3x+\frac{y}{\sqrt{3}}}\right)t_{12}\\+\left(S_{3x+\frac{2y}{\sqrt{3}}}-S_{3x-\frac{2y}{\sqrt{3}}}+S_{\frac{5x}{2}+\frac{7\sqrt{3}y}{6}}-S_{\frac{5x}{2}-\frac{7\sqrt{3}y}{6}}+S_{\frac{x}{2}-\frac{11\sqrt{3}y}{6}}-S_{\frac{x}{2}+\frac{11\sqrt{3}y}{6}}\right)t_{13}\end{pmatrix}. \tag{20}$$

Next, we provide the TDME, momentum matrix elements, and Berry curvature (BC) terms, calculated separately in both gauges. In gauge-I the x-components of the TDME take the form:

$$\begin{aligned}d^{\mathrm{I}}_{\mathrm{cv},x}(\mathbf{k})&=\frac{4\frac{\partial|\mathbf{B}|}{\partial k_x}\left(\mathrm{i}B_x+B_y\right)-2\mathrm{i}\left(\frac{\partial B_x}{\partial k_x}-\mathrm{i}\frac{\partial B_y}{\partial k_x}\right)\left(2|\mathbf{B}|+\Delta\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\\d^{\mathrm{I}}_{\mathrm{vv},x}(\mathbf{k})&=\frac{4\left(B_x\frac{\partial B_y}{\partial k_x}-B_y\frac{\partial B_x}{\partial k_x}\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\\d^{\mathrm{I}}_{\mathrm{cc},x}(\mathbf{k})&=\frac{-4\left(B_x\frac{\partial B_y}{\partial k_x}-B_y\frac{\partial B_x}{\partial k_x}\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\end{aligned} \tag{21}$$

where the forms for the y-components of the TDME are equivalent but where ∂_{k_x} are replaced by ∂_{k_y} terms, and the derivatives of $B_{x/y}$ can be readily obtained from the expressions above. Note that in gauge-I we have that $\mathbf{d}^{\mathrm{I}}_{\mathrm{vv}}(\mathbf{k})=-\mathbf{d}^{\mathrm{I}}_{\mathrm{cc}}(\mathbf{k})$. The expressions for the momentum matrix elements in gauge-I are:

$$\begin{aligned}p^{\mathrm{I}}_{\mathrm{cv},x}(\mathbf{k})&=\left(\frac{\partial B_x}{\partial k_x}-\mathrm{i}\frac{\partial B_y}{\partial k_x}\right)-\frac{8\left(B_x-\mathrm{i}B_y\right)\left(B_x\frac{\partial B_x}{\partial k_x}+B_y\frac{\partial B_y}{\partial k_x}\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\\p^{\mathrm{I}}_{\mathrm{vv},x}(\mathbf{k})&=\frac{\partial\alpha}{\partial k_x}-\frac{4\left(B_x\frac{\partial B_x}{\partial k_x}+B_y\frac{\partial B_y}{\partial k_x}\right)\left(2|\mathbf{B}|+\Delta\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\\p^{\mathrm{I}}_{\mathrm{cc},x}(\mathbf{k})&=\frac{\partial\alpha}{\partial k_x}+\frac{4\left(B_x\frac{\partial B_x}{\partial k_x}+B_y\frac{\partial B_y}{\partial k_x}\right)\left(2|\mathbf{B}|+\Delta\right)}{4|\mathbf{B}|^2+4B_x^2+4B_y^2+4|\mathbf{B}|\Delta+\Delta^2}\end{aligned} \tag{22}$$

where the y-components take a similar form with ∂_{k_x} replaced by ∂_{k_y}. The BC takes the form:

$$\Omega_{vv} = \frac{8\left(2\left|\mathbf{B}\right|+\Delta\right)\left(\begin{array}{c} 2\frac{\partial\left|\mathbf{B}\right|}{\partial k_y}\left(B_x\frac{\partial B_y}{\partial k_x}-B_y\frac{\partial B_x}{\partial k_x}\right)+2\frac{\partial\left|\mathbf{B}\right|}{\partial k_x}\left(B_y\frac{\partial B_x}{\partial k_y}-B_x\frac{\partial B_y}{\partial k_y}\right) \\ -\left(\frac{\partial B_x}{\partial k_y}\frac{\partial B_y}{\partial k_x}-\frac{\partial B_x}{\partial k_x}\frac{\partial B_y}{\partial k_y}\right)\left(2\left|\mathbf{B}\right|+\Delta\right)\end{array}\right)}{\left(4\left|\mathbf{B}\right|^2+4B_x^2+4B_y^2+4\left|\mathbf{B}\right|\Delta+\Delta^2\right)^2} \tag{23}$$

and $\Omega_{cc}=-\Omega_{vv}$. Note that in gauge-I there is a singularity in the expression for Ω for the case of $\Delta=0$. Regardless, for $\Delta=0$ one should identically have $\Omega\equiv 0$ since the system is inversion symmetric.

Expressions for the TDME and momentum matrix elements in gauge-II are given below for the specific case of $\Delta=0$, which is simplified, and the one employed in the main text:

$$\begin{aligned} d^{\mathrm{II}}_{\mathrm{cv},x}\left(\mathbf{k}\right) &= \frac{B_x\frac{\partial B_y}{\partial k_x}-B_y\frac{\partial B_x}{\partial k_x}}{2\left|\mathbf{B}\right|^2} \\ d^{\mathrm{II}}_{\mathrm{vv},x}\left(\mathbf{k}\right) &= \frac{B_y\frac{\partial B_x}{\partial k_x}-B_x\frac{\partial B_y}{\partial k_x}}{2\left|\mathbf{B}\right|^2} \\ d^{\mathrm{II}}_{\mathrm{cc},x}\left(\mathbf{k}\right) &= \frac{B_y\frac{\partial B_x}{\partial k_x}-B_x\frac{\partial B_y}{\partial k_x}}{2\left|\mathbf{B}\right|^2} \end{aligned} \tag{24}$$

with expressions for the y-components being equivalent with ∂_{k_x} terms replaced by ∂_{k_y}. Note that in equation (24) $\left|\mathbf{B}\right|^2=B_x^2+B_y^2$ due to the gap term vanishing, and that in gauge II for $\Delta=0$ we have that $\mathbf{d}^{\mathbf{II}}_{\mathbf{cv}}=-\mathbf{d}^{\mathbf{II}}_{\mathbf{vv}}$, and $\mathbf{d}^{\mathbf{II}}_{\mathbf{vv}}=\mathbf{d}^{\mathbf{II}}_{\mathbf{cc}}$, which simplifies the SBE. We further note that at K/K' points the TDME diverge for like $1/\left|\mathbf{k}\right|$, but numerically we found that this divergence is more manageable than expressions in gauge I. The momentum matrix elements in gauge-II for $\Delta=0$ are:

$$\begin{aligned} p^{\mathrm{I}}_{\mathrm{cv},x}\left(\mathbf{k}\right) &= i\frac{B_x\frac{\partial B_y}{\partial k_x}-B_y\frac{\partial B_x}{\partial k_x}}{\left|\mathbf{B}\right|} \\ p^{\mathrm{I}}_{\mathrm{vv},x}\left(\mathbf{k}\right) &= \frac{\partial\alpha}{\partial k_x}-\frac{B_x\frac{\partial B_x}{\partial k_x}+B_y\frac{\partial B_y}{\partial k_x}}{\left|\mathbf{B}\right|} \\ p^{\mathrm{I}}_{\mathrm{cc},x}\left(\mathbf{k}\right) &= \frac{\partial\alpha}{\partial k_x}+\frac{B_x\frac{\partial B_x}{\partial k_x}+B_y\frac{\partial B_y}{\partial k_x}}{\left|\mathbf{B}\right|} \end{aligned} \tag{25}$$

with similar expressions for the y-components but with ∂_{k_x} terms replaced by ∂_{k_y}.

Table 2. Values for optimally fitted hopping amplitudes (t_j, in eV), gaps (Δ , in eV), and lattice parameters (a_0, in $\mathring{A}$) employed in the 14th-NN TB model, fitted throughout the entire BZ with double weight near K/K' regions.

	graphene	hBN	MoS_2	$MoSe_2$	$MoTe_2$	WS_2	WSe_2	WTe_2
a_0	2.460	2.520	3.184	3.320	3.547	3.186	3.319	3.550
Δ	0	4.42694	1.71162	1.477 06	1.11918	1.79675	1.53674	1.06764
t_1	−1.98019	−1.67609	0.54087	0.48313	−0.44597	−0.58442	0.52326	−0.48832
t_2	0.01124	−0.08150	0.02462	0.01565	0.02988	0.00524	−0.00142	0.01993
t_3	−0.10678	−0.05289	−0.28305	−0.21477	0.11580	0.38689	−0.30580	0.19528
t_4	0.58587	0.52941	0.04214	0.07181	−0.08566	−0.07450	0.09874	−0.10715
t_5	−0.00263	−0.00258	0.05191	0.05225	0.04864	0.05345	0.05304	0.04748
t_6	−0.00262	−0.02598	0.03753	0.03858	0.04049	0.04899	0.04607	0.04299
t_7	−0.22366	−0.15549	−0.02551	−0.02607	0.01568	0.02207	−0.02069	0.01504
t_8	0.24277	0.17971	0.11495	0.11591	−0.09806	−0.12618	0.12871	−0.11781
t_9	0.05666	0.04557	0.00216	0.00371	0.00207	0.00072	−0.00002	0.00575
t_{10}	0.02043	0.02183	0.00050	0.00507	0.00936	−0.00002	0.00507	0.01138
t_{11}	0.01833	0.01789	−0.00896	−0.00433	−0.00134	−0.01188	−0.00761	−0.00548
t_{12}	0.04847	0.03999	−0.00691	−0.00764	0.00561	0.01521	−0.01597	0.01428
t_{13}	−0.07728	−0.07020	−0.00072	−0.00051	0.00134	0.00219	−0.00011	−0.00236
t_{14}	0.00112	0.00149	−0.00702	−0.00582	−0.00253	−0.00797	−0.00637	−0.00246

Table 3. Values for optimally fitted hopping amplitudes (t_j, in eV) employed in the 2nd-NN TB model, which were fitted only near the K/K' regions. The gap and lattice parameters in each system are the same as in table 2 above.

	graphene	hBN	MoS_2	$MoSe_2$	$MoTe_2$	WS_2	WSe_2	WTe_2
t_1	2.00000	1.99945	−1.14057	0.93764	0.72098	1.48144	1.23622	−0.97070
t_2	−0.36730	−0.26244	0.05685	0.03015	0.04466	0.15882	0.10331	0.11474

Tables 2 and 3 below provide the optimal values of hopping parameters, gaps, and lattice parameters used for each material system (for the 14th NN and 2nd NN cases, respectively). Values are provided in eV.

Appendix B. Additional data

We present here complementary data to that in the main text. First, for completeness figure 6 compares the conduction band BC in hBN in the 2NN and 14NN models. Clearly, minor changes to the BC are expected since by adding more hopping terms the band structure itself changes, which follows changes in the eignestates of the TB Hamiltonian and resulting BC. However, we find these changes relatively minute such that 2nd order TB models have a pretty accurate BC. The main differences are that the long-range model captures stronger trigonal wrapping effects, slightly increases the peak value of the BC at K/K' (by ∼20%, perhaps modulating yields of even harmonics as discussed in the main text), and causes minor BC contributions away from K/K' (typically ∼10% compared to main peak).

Second, figures 7–14 below present the full angular dependence of HHG yields per harmonic order in each system in similar conditions to those discussed in the main text, where the material system in each case is indicated in inset. This complements results discussed in the main text for select harmonics.

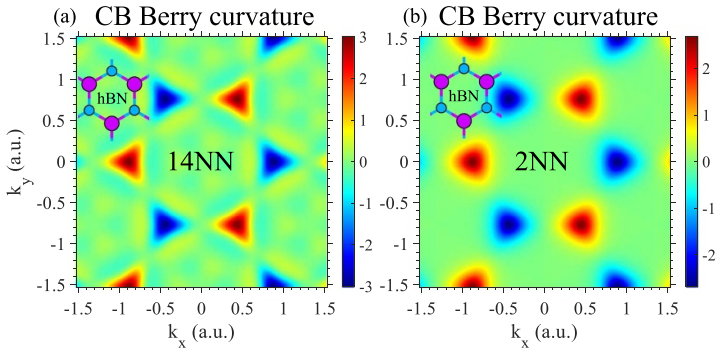


Figure 6. Difference in Berry curvature in the CB of hBN for different order TB models (a) 14NN, (b) 2NN.

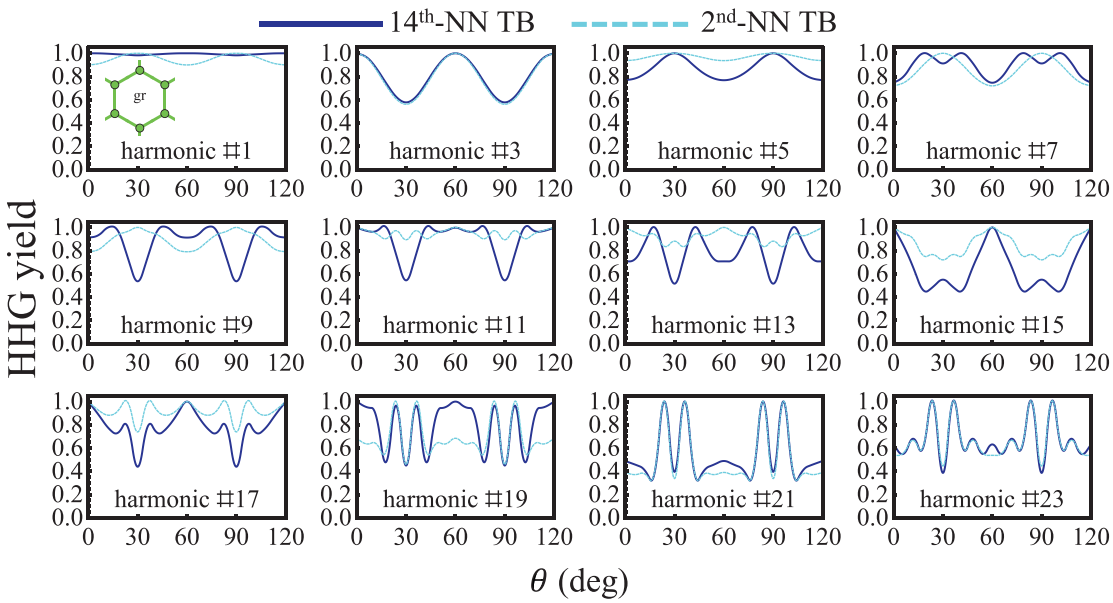


Figure 7. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from graphene.

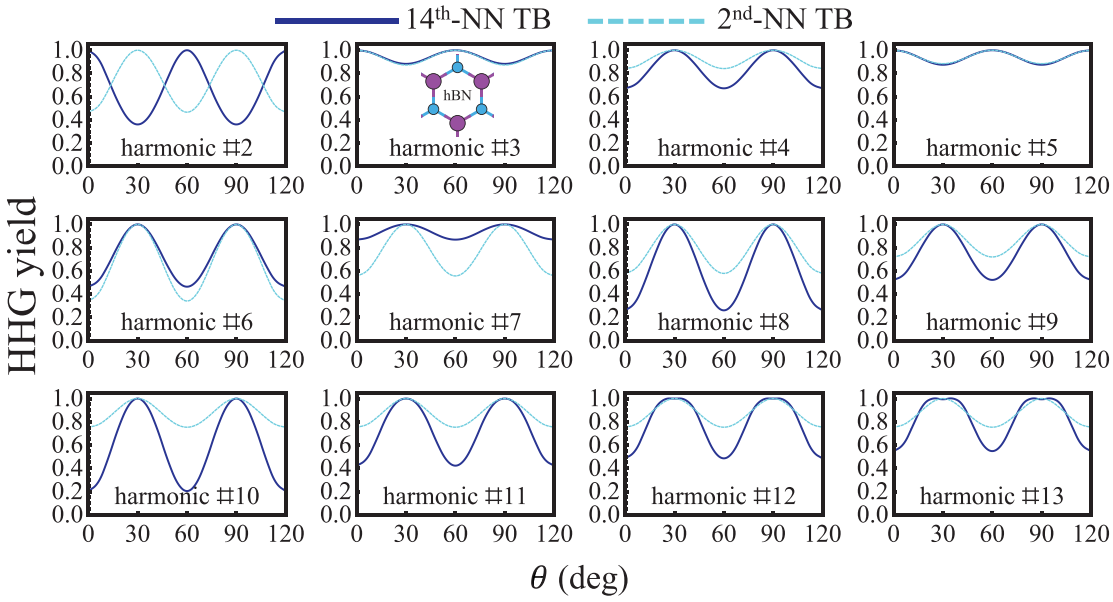


Figure 8. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from hBN.

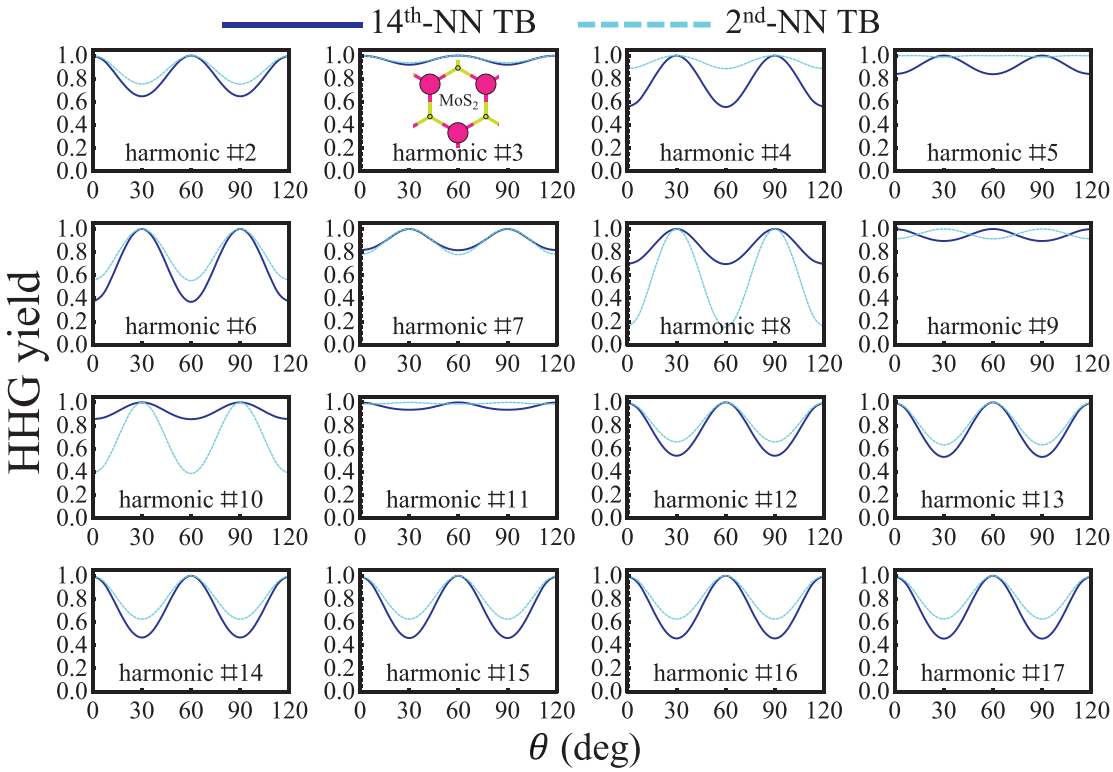


Figure 9. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from MoS_2.

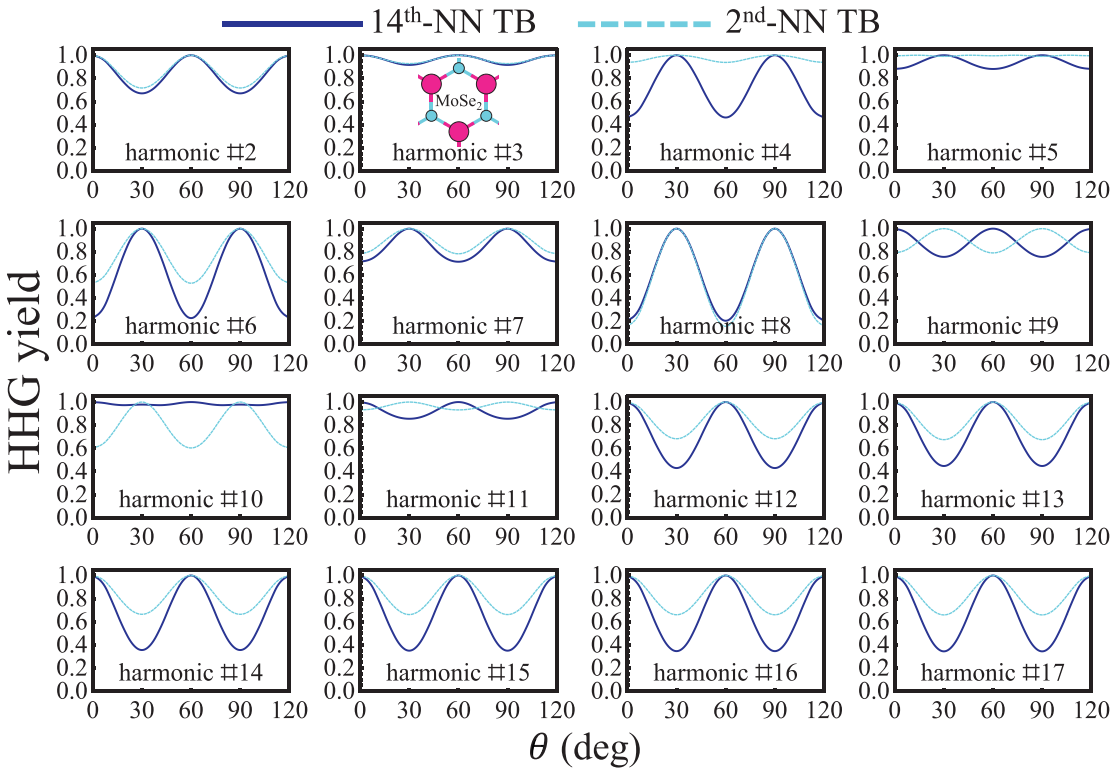


Figure 10. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from $MoSe_2$.

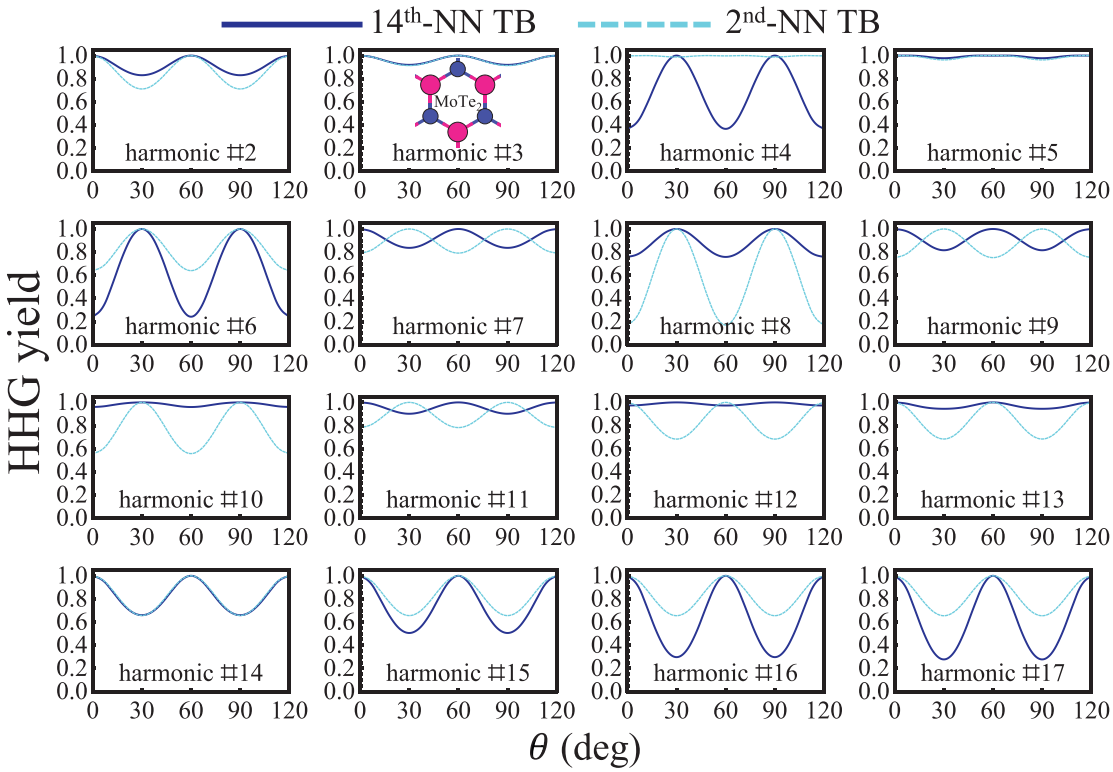


Figure 11. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from $MoTe_2$.

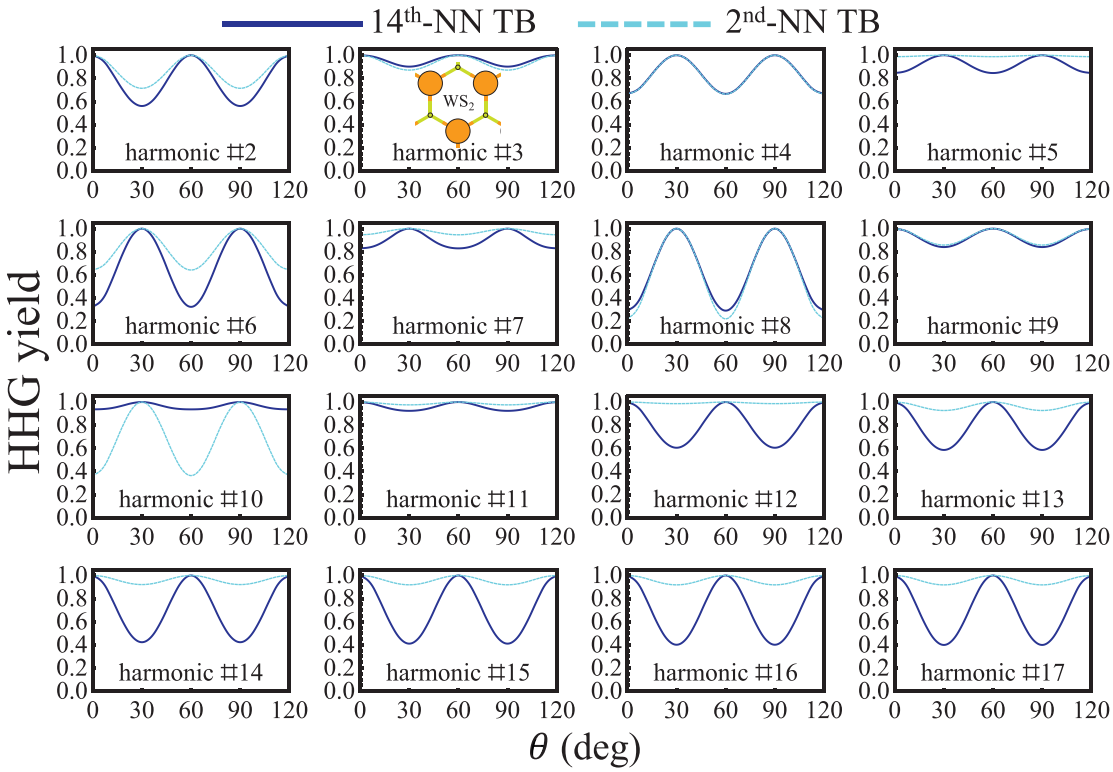


Figure 12. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from WS_2.

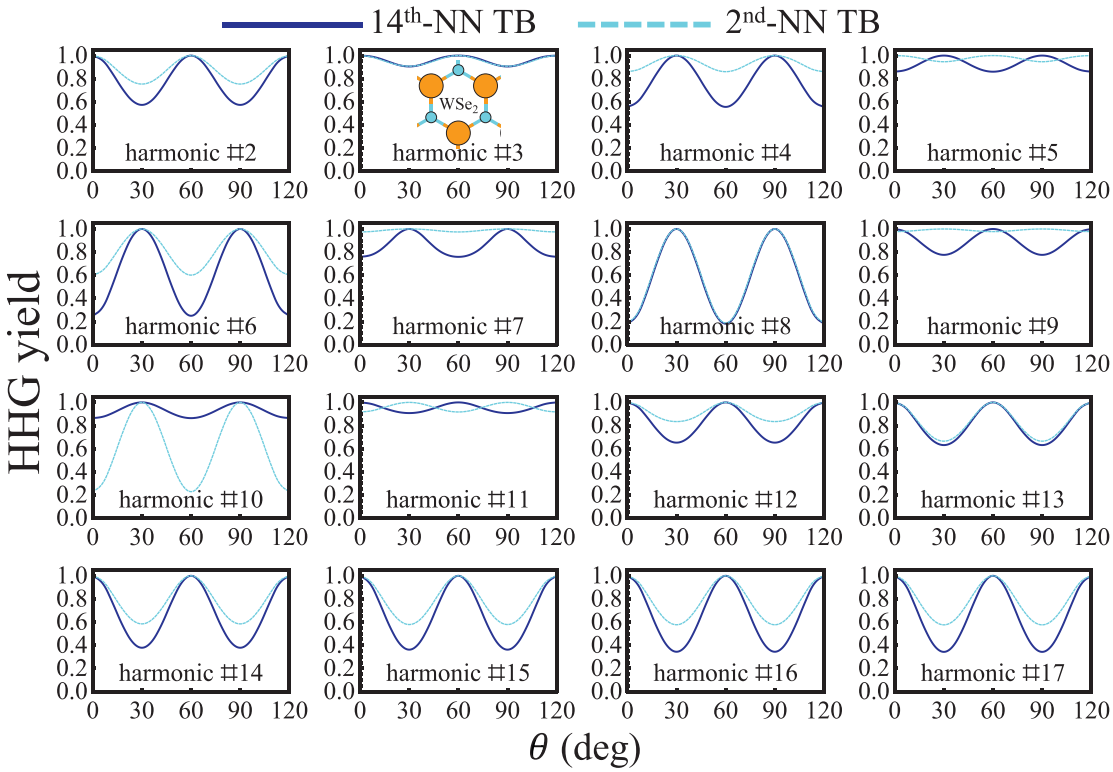


Figure 13. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from WSe_2.

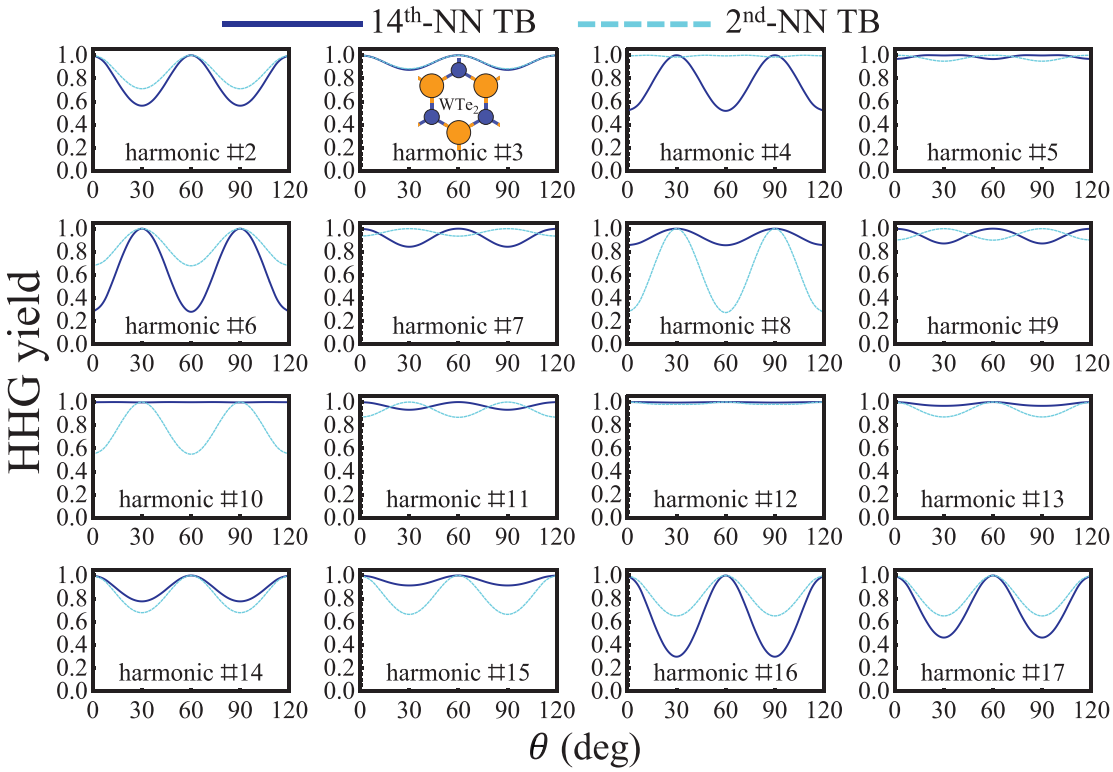


Figure 14. Angle-resolved normalized harmonic yields for the 14th-NN TB model in similar conditions to figures 2–3 in the main text from WTe_2.

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